

3 - Deep Learning

FFNN / FCNN / ANXI / Dense

$$y_{act} - y_{pred.}$$



(error)

x. کون کس طرح سے ملے گی؟

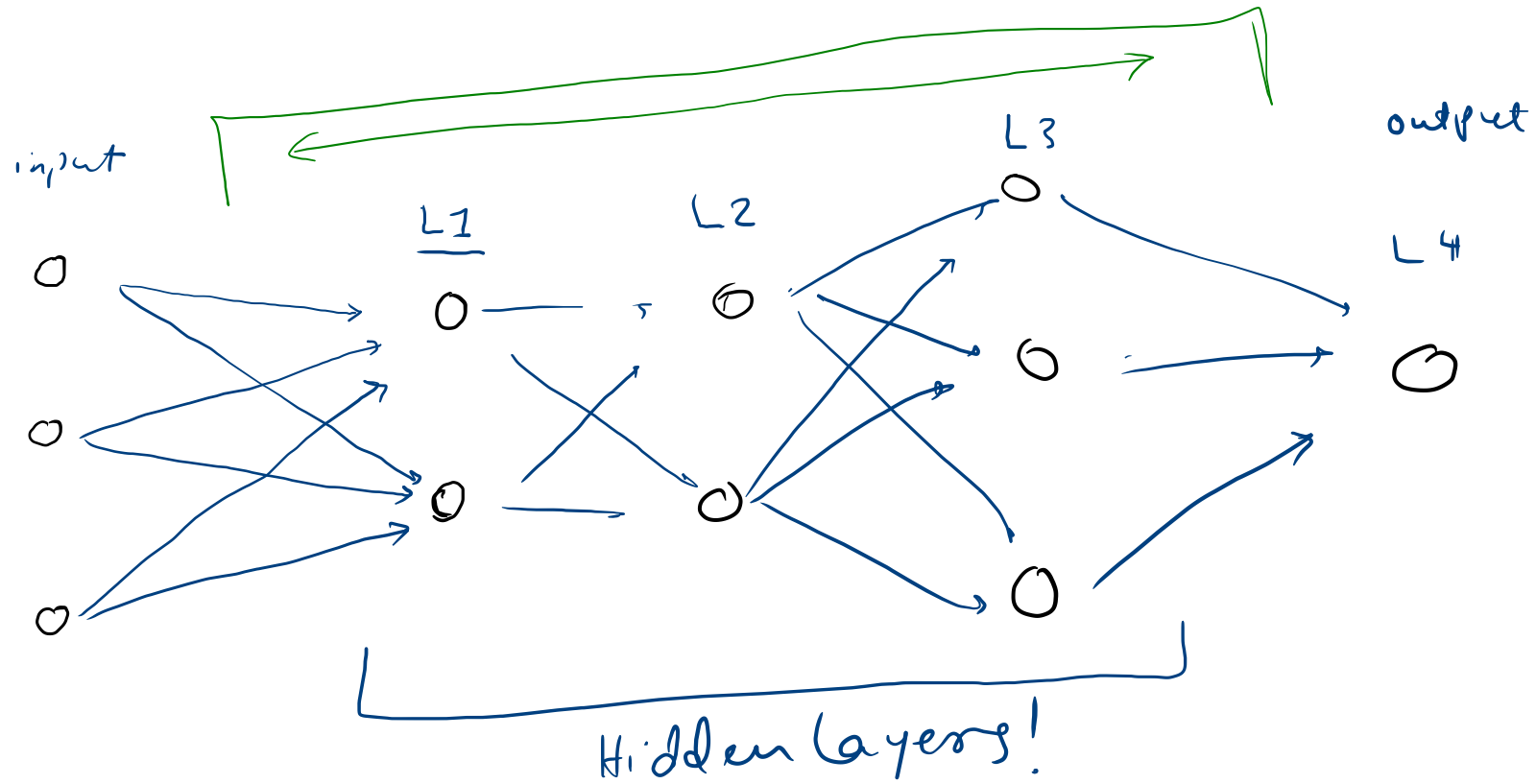


تعیین وزن، w ، بایس (b)، درستی/غلطی (error)

(loss f.) تابع زیان
(cost f.) تابع هزینه

- MSE
- MAE
- CE
- BCE

چیزهایی؟
 کسی؟
 هر حرفی که بعضی
 می توانند صحنه زعفرانی
 به عنوان ورودی
 داشته باشند



این حرفی که بعضی هستند که ورودی دارند؟
 آری می توانند صحنه زعفرانی داشته باشند؟
 بله

x House price estimation

f1	f2 ^a	Target
area	age	price

Model Designing
طراحی مدل

کدام ورودی ها را می توانیم داشته باشیم ؟
کدام خروجی ها را می توانیم داشته باشیم ؟

input

output

number of input neurons = 2
" " output neurons = 1



ساده

Dataset

Dogs

→ 10K

Cats

→ 11K

image

Deep Model

output

Dog

or ?

Cat

داده های تصویری چگونه به مدل می رسد؟

فرآیند تبدیل تصویر به بردار چگونه است؟

Image

Image

⑦

0	8	3	9	0
14	8	13	3	3
5	5	6	2	3
10	16	18	2	56

4px

①

matrix → vector

2d → 1d

0	f1
8	f2
3	f3
9	
⋮	
16	
18	
255	f16

1x16

0
0
0
0
⋮
0

α

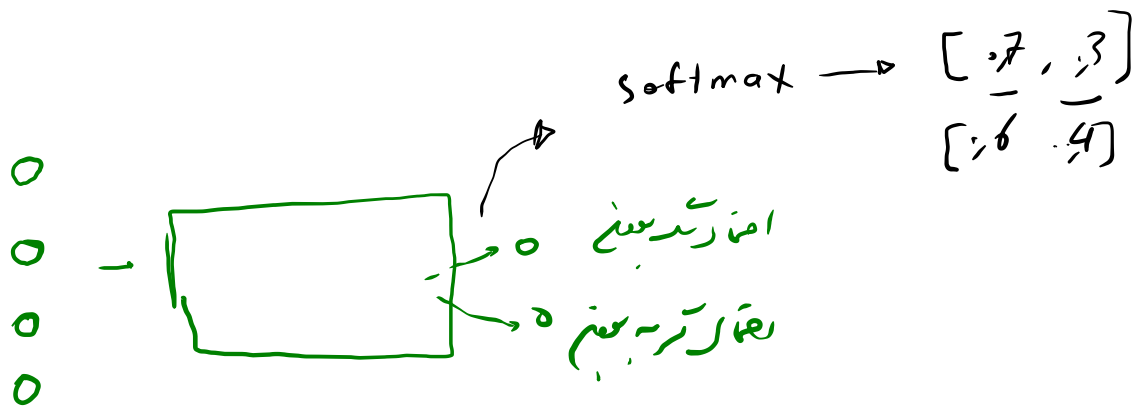
output

احتمال گربه بودن
احتمال سگ بودن

مثال: $\begin{bmatrix} 0.7 \\ 0.3 \end{bmatrix}$
احتمال گربه بودن: 0.7
احتمال سگ بودن: 0.3

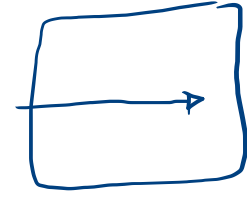
چگونه تصویر را به بردار تبدیل می کنند؟
هر پیکسل در تصویر به یک عدد تبدیل می شود.

عوامل				Target
F1	F2	F3	F4	تدريج
اندازه کوس	اندازه د	قد	وزن	



input : مقدار درجہ

0
0
0
0
0



output : ①
0

Regression

خرید و فروش سیرتے ہیں!

- مقدار درجہ
- عمر کے ساتھ
- مہنگائی

classification $\rightarrow$ $\left\{ \begin{array}{l} \text{input} \\ \text{نمبر، رنگ، وزن} \end{array} \right\} \rightarrow \left\{ \begin{array}{l} \text{output} \\ \text{نمبر، رنگ} \end{array} \right\}$

سوال

Dataset : $h \overset{w}{\boxed{\text{dog}}} \leftarrow \underline{100K}$

Model

سوال : مفهوم در درستی یا اشتباهی باشد

فکر کرد در درستی یا اشتباهی باشد

$w \times h$



$\frac{1}{\text{Sigmoid}}$

احتمال درستی !

نویسندگی

Activation Func.

$$\text{Sigmoid} = \sigma = \frac{1}{1 + e^{-x}}$$

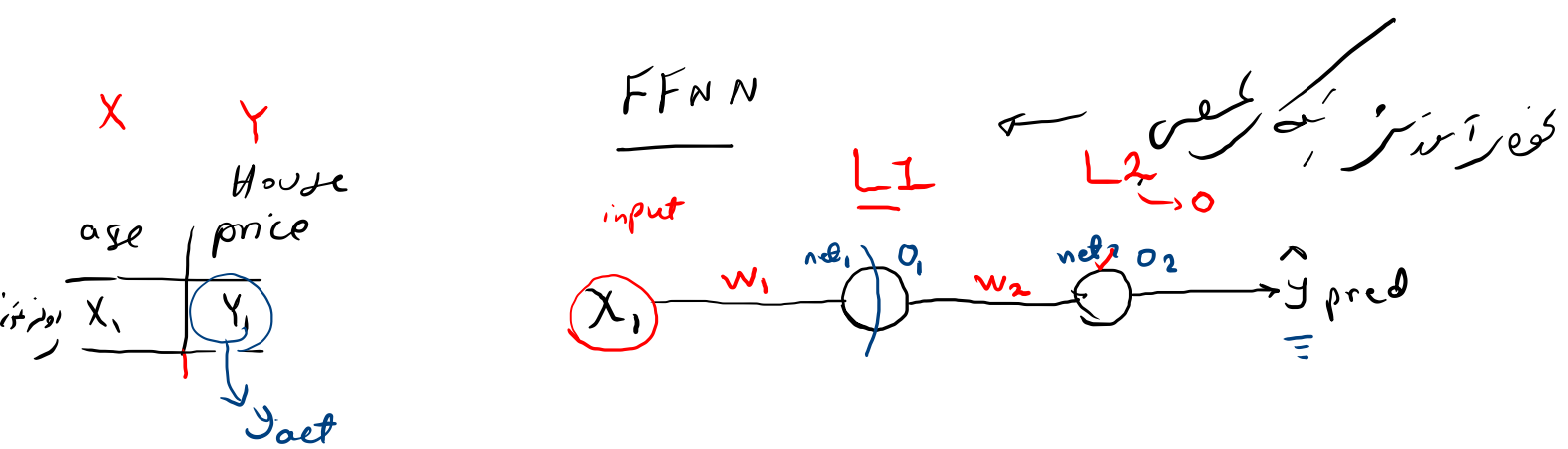
$$\text{Softmax} = \frac{e^{x_i}}{\sum_{i=1}^n e^{x_i}}$$

فرم زیر از عدد از softmax

$$\rightarrow \begin{bmatrix} 1.3 \\ 9 \\ 11 \end{bmatrix}$$

$$\begin{bmatrix} \frac{e^{1.3}}{e^{1.3} + e^9 + e^{11}} \\ \frac{e^9}{e^{1.3} + e^9 + e^{11}} \\ \frac{e^{11}}{e^{1.3} + e^9 + e^{11}} \end{bmatrix} = \begin{bmatrix} .07 \\ .4 \\ .53 \end{bmatrix}$$

→ $\square$ → μ → $\begin{bmatrix} 1.3 \\ 9 \\ 11 \end{bmatrix}$ → softmax → $\begin{bmatrix} .07 \\ .4 \\ .53 \end{bmatrix}$



Regression → Loss function: MSE



Gradient Descent
کراسش به سمت پایین

$$w_{new} = w_{old} - \alpha \frac{\partial L}{\partial w_{old}}$$

Handwritten notes and calculations for Gradient Descent:

- $w_1 = 0$
- Iteration 1: $w = 0 - \alpha \frac{\partial L}{\partial w}$ → -0.3
- Iteration 2: $w = -0.3 - \alpha \frac{\partial L}{\partial w}$ → 1
- Iteration 3: $w = 1 - \alpha \frac{\partial L}{\partial w}$ → 1.7

Final result: $w = 1.7$

$$y \rightarrow x \rightarrow u$$

سوال

$$y = x^2 + 4x$$

$$x = 2u + 1$$

$$\frac{dy}{du}$$

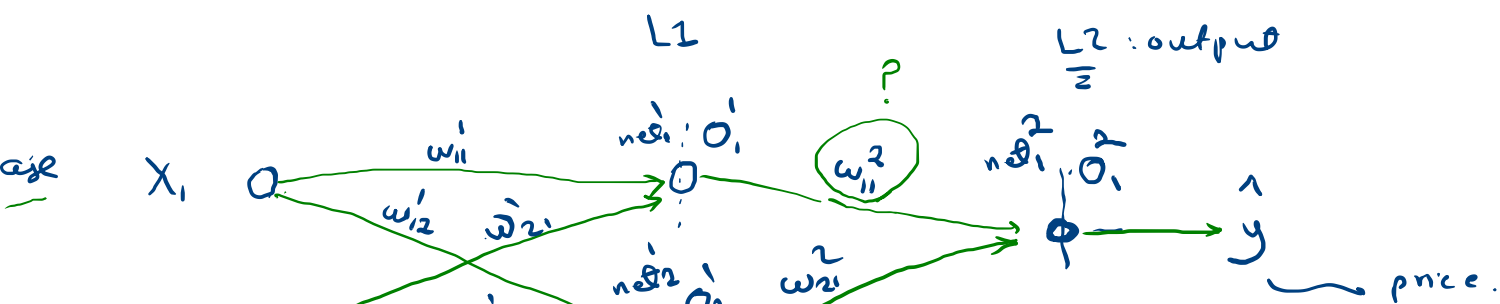
$$= \frac{\partial y}{\partial x} \cdot \frac{\partial x}{\partial u}$$

$$\frac{\partial x}{\partial u} = 2$$

$$= [2x + 4](2) = 4x + 8$$

$$\frac{\partial y}{\partial x} = 2x + 4$$

سوال آخری



$b_i = 0$

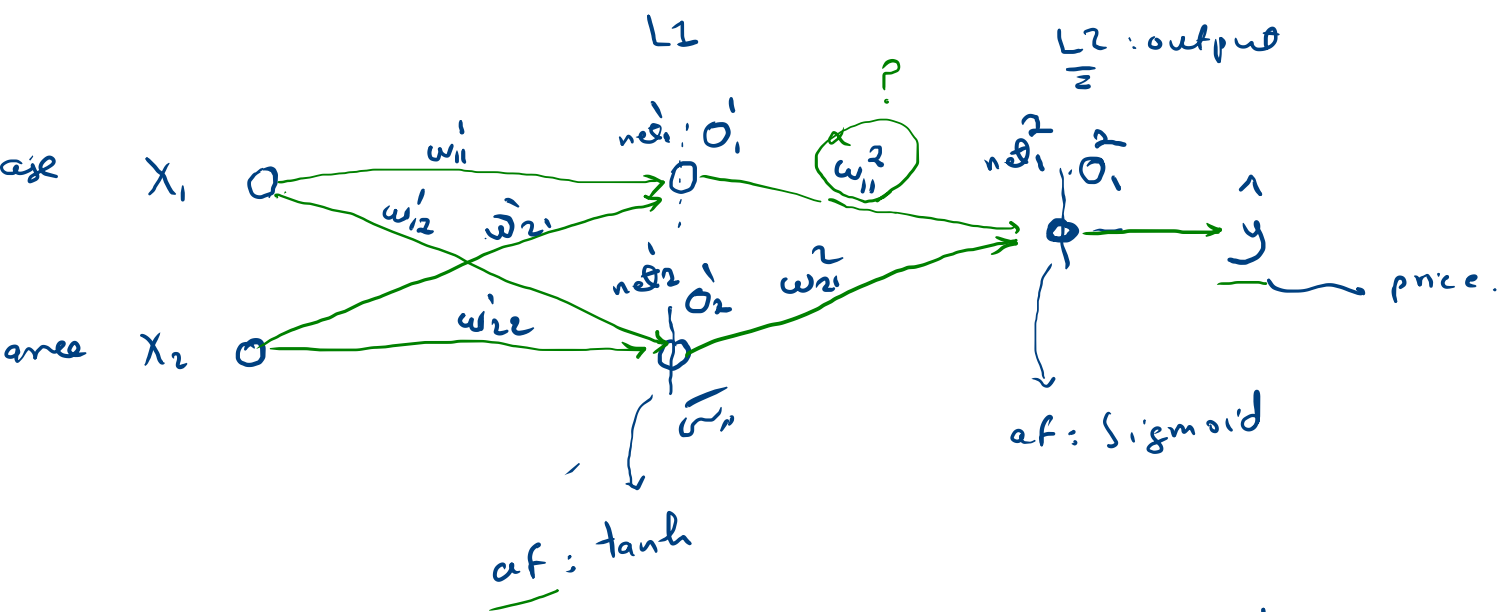
GD (Gradient Descent) update rule for weight w_{11}^2 :

$$w_{11}^2 - \text{new} = w_{11}^2 - \alpha \frac{\partial L}{\partial w_{11}^2}$$

Where α is the learning rate.

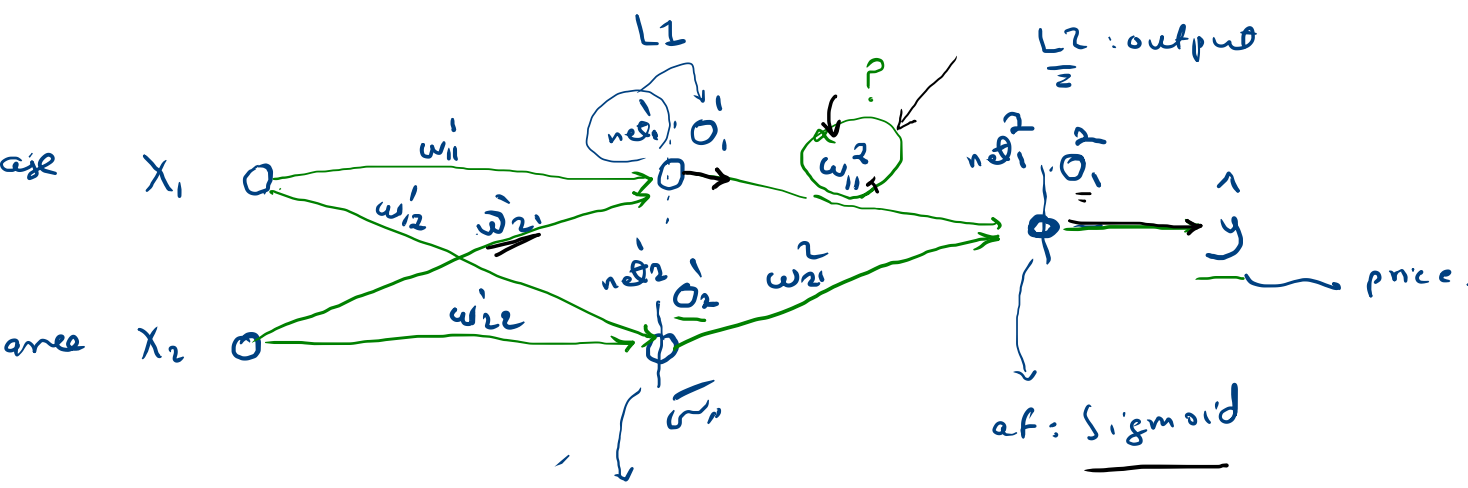
Randomly: w_{11}^2

Derivative: $\frac{\partial L}{\partial w_{11}^2} = ?$



$$w_{11}^2 - new = ?$$

The End



$b_i = 0$ وضوح

L1 {

$$\begin{aligned} \text{net}_1' &= w_{11}' x_1 + w_{21}' x_2 \\ \text{net}_2' &= w_{12}' x_1 + w_{22}' x_2 \end{aligned}$$

$\rightarrow \underline{o_1'} = \tanh(\text{net}_1')$

$\rightarrow \underline{o_2'} = \tanh(\text{net}_2')$

new $\omega_{11}^2 = \omega_{11}^2 - \alpha \frac{\partial L}{\partial \omega_{11}^2}$ Hyper parameter.

$(L) = \text{mse} = \frac{1}{n} \sum (y_{\text{act}} - y_{\text{pred}})^2$

$\frac{\partial L}{\partial \omega_{11}^2} = \frac{\partial L}{\partial y_{\text{pred}}} \frac{\partial y_{\text{pred}}}{\partial o_1^2} \frac{\partial o_1^2}{\partial \text{net}_1^2} \frac{\partial \text{net}_1^2}{\partial \omega_{11}^2}$

$\text{net}_1^2 = w_{11}^2 o_1' + w_{21}^2 o_2'$

$\underline{o_1^2} = \sigma(\text{net}_1^2) = \hat{y}_{\text{pred}}$

$\frac{\partial y_{\text{pred}}}{\partial o_1^2} = 1$

$\frac{\partial o_1^2}{\partial \text{net}_1^2} = \sigma'(\text{net}_1^2) = \sigma(\text{net}_1^2)(1 - \sigma(\text{net}_1^2))$

$$A: \frac{\partial L}{\partial y_p} = \frac{1}{N} (2)(y_{\text{act}} - y_p)(-1) = \frac{-2}{N} (y_{\text{act}} - y_p)$$

$$\frac{\partial \sigma_1}{\partial \text{net}_1} = ?$$

$$\sigma_1 = \delta(\text{net}_1) \Rightarrow \frac{\partial \sigma_1}{\partial \text{net}_1} = \frac{\partial \sigma_1}{\partial \delta} \left[\frac{\partial \delta}{\partial \text{net}_1} \right]$$

$$\delta = \frac{1}{1 + e^{-x}}$$

$\text{net}_1^2 = n$

$$\frac{\partial \delta}{\partial \text{net}_1^2} = \frac{\partial}{\partial \text{net}_1^2} \left[\frac{1}{1 + e^{-\text{net}_1^2}} \right] = \frac{\partial}{\partial \text{net}_1^2} (1 + e^{-n})^{-1} = -1 (1 + e^{-n})^{-2} (-e^{-n})$$

$$\frac{\partial}{\partial x} \left(\frac{1}{x} \right) = -\frac{1}{x^2} \rightarrow \left(\frac{v}{v} \right)' = \frac{v'v - v'v}{v^2} \Rightarrow \left(\frac{1}{v} \right)' = \frac{-v'v}{v^2} = -\frac{v'}{v^2}$$

$$v = 1 + e^{-x} \rightarrow \left(\frac{1}{v} \right)' = \frac{-(-e^{-n})}{(1 + e^{-n})^2} = \frac{e^{-n}}{(1 + e^{-n})^2}$$

vir p

$$\frac{\partial^2 \sigma_1}{\partial \text{net}_1^2} = \frac{\partial}{\partial \text{net}_1} \left(\frac{1}{1 + e^{-\text{net}_1}} \right) = \frac{e^{-\text{net}_1}}{(1 + e^{-\text{net}_1})^2} = \frac{+1 - 1 + e^{-\text{net}_1}}{(1 + e^{-\text{net}_1})^2}$$

$$\frac{\partial^2 \sigma_1}{\partial \text{net}_1^2} = \frac{-1 + (1 + e^{-\text{net}_1})}{(1 + e^{-\text{net}_1})^2} = \boxed{\frac{-1}{(1 + e^{-\text{net}_1})^2}} + \boxed{\frac{1}{(1 + e^{-\text{net}_1})^2}}$$

$$\frac{\partial^2 \sigma_1}{\partial \text{net}_1^2} = -b^2 + b \rightarrow \frac{\partial^2 \sigma_1}{\partial \text{net}_1^2} = b(n) (1 - b(n))$$

$b(n) = \frac{1}{1 + e^{-\text{net}_1}}$

$$\begin{aligned} \sigma_1 &= b(n) \\ \frac{\partial \sigma_1}{\partial x} &= b(n)(1 - b(n)) \end{aligned}$$

$$v = u(x)^2$$



$$\frac{u}{x} = ?$$



$$\frac{x}{u} \left(\frac{u}{x} \right)$$

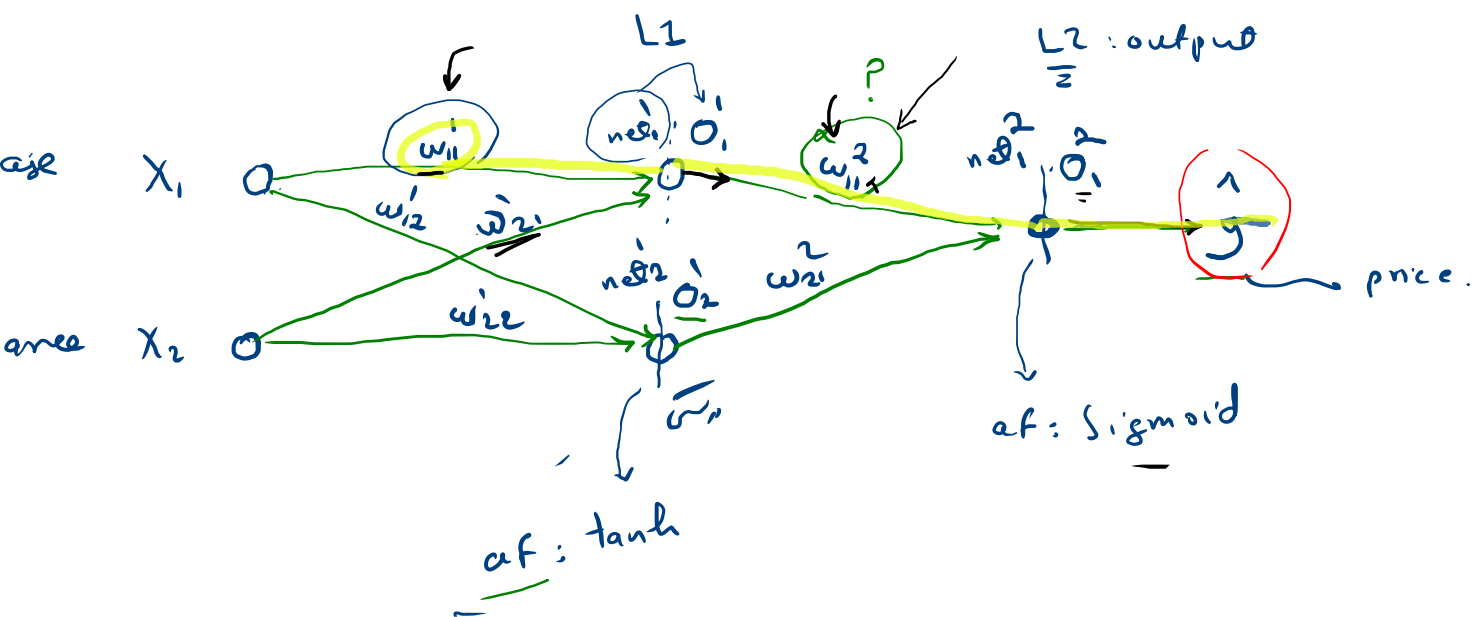
$$= (x - u)^2$$

$$x - u = 0$$

$$L = \sum (y - \hat{y}_x)^2 \Rightarrow L = \sum^2 \Rightarrow \frac{\partial L}{\partial y_x} = 2y = \frac{\partial L}{\partial y_x} = 2y = 2(y - \hat{y}_x) \Rightarrow \frac{\partial L}{\partial y_x} = 2(y - \hat{y}_x)$$

$$\frac{\partial \mathcal{L}}{\partial w_{11}^2} = - \underbrace{\frac{2}{N}}_{\checkmark} \left(\underbrace{y_{\text{act}}}_{\checkmark} - \underbrace{y_{\text{pred}}}_{\checkmark} \right) (1) \underbrace{\delta(\text{net}_i^2)}_{\checkmark} \underbrace{(1 - \delta(\text{net}_i^2))}_{\checkmark} \underbrace{O_i}_{\checkmark}$$

$$\underbrace{w_{11}^2}_{\text{new}} = w_{11}^2 - \alpha \frac{\partial \mathcal{L}}{\partial w_{11}}$$



$b_i = 0$ (circled in red)

L1 {

$$net_1' = \underline{w_{11}'} x_1 + \underline{w_{21}'} x_2$$

$$o_1' = \tanh(net_1')$$

$$net_2' = w_{12} x_1 + w_{22} x_2$$

$$o_2' = \tanh(net_2')$$

}

L2 {

$$net_1^2 = \underline{w_{11}^2} o_1' + w_{21}^2 o_2'$$

$$o_1^2 = \delta(net_1^2) = \hat{y}_{pred}$$

}

$\hat{y}_{pred} \rightarrow w_{11}' = w_{11}' - \alpha \left(\frac{\partial L}{\partial w_{11}'} \right)$

$$\frac{\partial L}{\partial w_{11}} = \frac{\partial L}{\partial \hat{y}_{pred}} \frac{\partial \hat{y}_{pred}}{\partial o_1^2} \frac{\partial o_1^2}{\partial net_1^2} \frac{\partial net_1^2}{\partial o_1'} \frac{\partial o_1'}{\partial net_1'} \frac{\partial net_1'}{\partial w_{11}'}$$

age	area	price
3	30	3
4	90	5
50	120	5



3. Loss function: $\text{mse} = \frac{1}{3} \sum_i (y_{\text{act}} - y_{\text{pred}})^2$

$$\text{mse} = \frac{1}{3} [(3-4)^2 + (5-4.5)^2 + (5-5.7)^2]$$

4.

$$w_{ij}^{\text{new}} = w_{ij} - \alpha \frac{\partial L}{\partial w_{ij}}$$

$$b_i^{\text{new}} = b_i - \alpha \frac{\partial L}{\partial b_i}$$

$\vdots$

$\rightarrow 5: \text{price}$

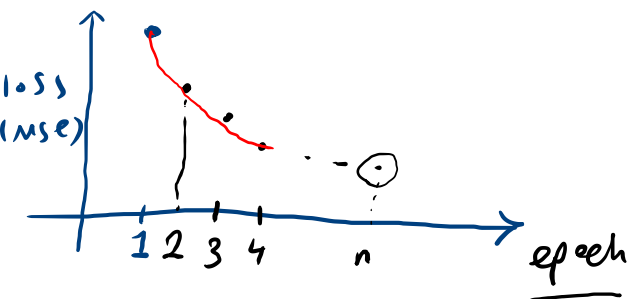
1. $w, b \rightarrow$ randomly initialized

2. $s1 \left\{ \begin{array}{l} 3 \\ 30 \end{array} \right\} \rightarrow y_{\text{pred}1} : 4$

$s2 \left\{ \begin{array}{l} 4 \\ 90 \end{array} \right\} \rightarrow y_{\text{pred}2} : 4.5$

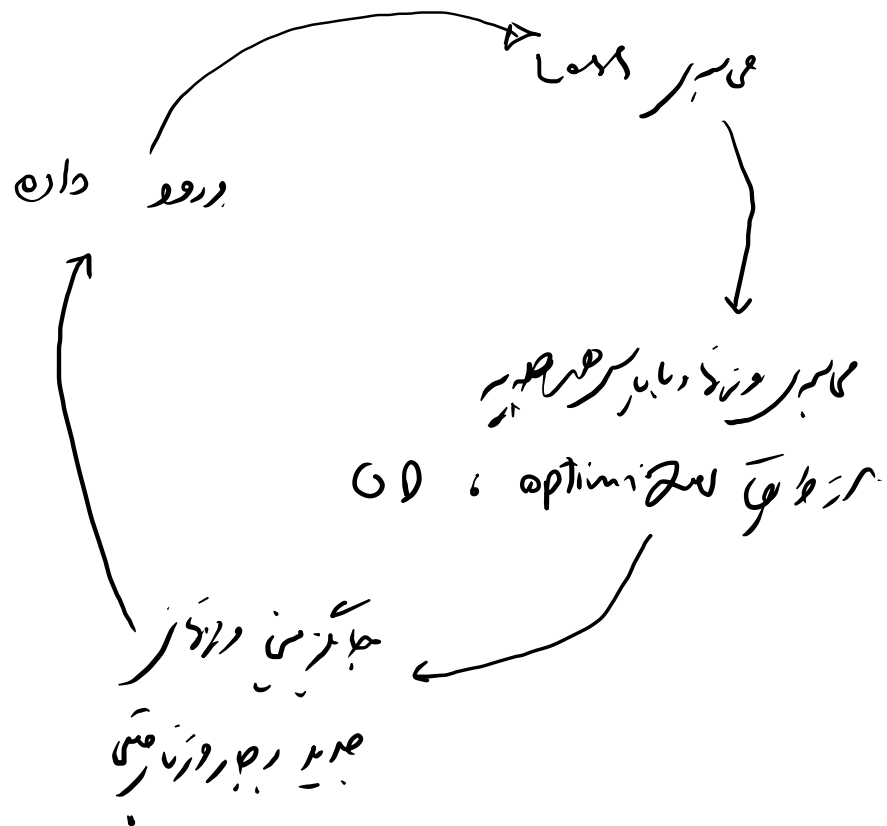
$s3 \left\{ \begin{array}{l} 50 \\ 120 \end{array} \right\} \rightarrow y_{\text{pred}3} : 5.7$

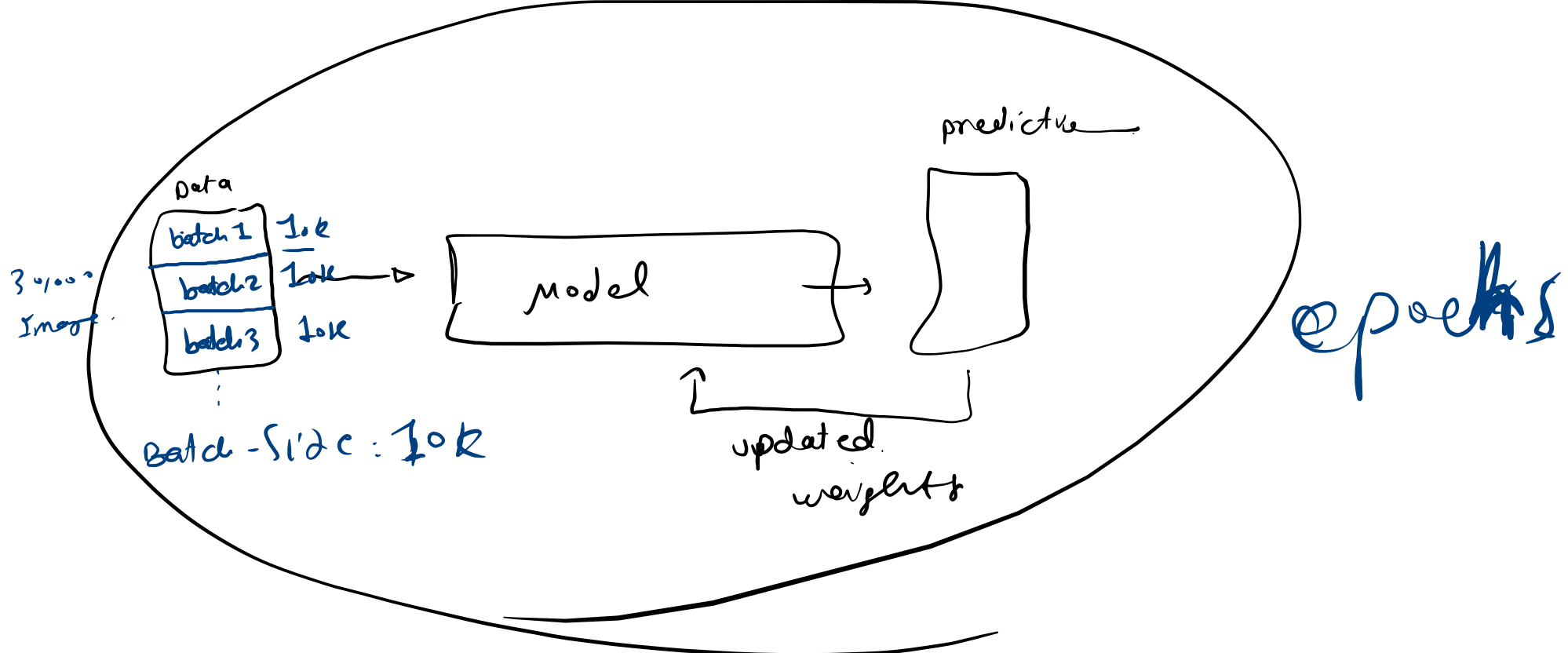
$\vdots$



epoch : تعداد دفعات

شبکه عصبی، تمام داده‌ها
آموخته می‌شود، هدف ما
است!





epoch {

- Batch 1 → model → w. update
- Batch 2 → model → w. update
- Batch 3 → model → w. update